

claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\agent-a89ebe7a124c60412.jsonl`  
- SHA-256: `84376e76c2b0bd9ef14f096e2ebd546f1c357cf21995370d035d2f1303959662`  
- Source modified: `2026-06-18T08:14:45+00:00`  
- Imported at: `2026-07-05T16:48:22+00:00`  
- project: `subagents`  
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`
```

Transcript

1. user (2026-06-18T08:12:18.181Z)

Fix the 3 stale assertions in `tests/test\_served\_gate\_a\_regression.py` in the badminton-highlight-indexer repo (you are in an isolated git worktree of it). They FAIL locally because the golden corpus grew from the original 6 videos to 15 (9 currently have both trajectory + label files present), while the test hardcodes $n=6$. They are `cv2`-gated (`pytest.importorskip("cv2")`) so they SKIP in CI and only fail on a machine where the golden data is present.

CONTEXT: this is the SERVED Gate-A litmus (the #146 follow-up) ? an INDEPENDENT track from the long-rally lens (PR #220, do NOT touch it). One PR per work item.

GOLDEN DATA (critical): you are in an isolated worktree, so the gitignored `output/` golden files are NOT in your worktree. The manifest + golden data live in the MAIN checkout. The test supports an env override ? set:

```
export SERVED_GATE_A_MANIFEST="C:/Users/avidu/Projects/badminton-highlight-indexer/output/human_lovo_manifest.json"
```

The manifest's `trajectory` and `labels` fields are ABSOLUTE paths (into the main checkout / "Annotation Setup"), so with that env var set the test resolves the data from your worktree. (You are on Windows; the Bash tool is Git Bash ? `export VAR=...; python ...` works.)

STEP 0 ? REPRODUCE OR STOP: with the env var set, run `python -m pytest tests/test\_served\_gate\_a\_regression.py -q`. You MUST see the 3 failures (NOT "skipped"). If they SKIP (golden data not visible from the worktree), STOP and report ? do NOT "fix" blind.

THE 3 FIXES ? fix the test's INTENT; do NOT merely widen bands to force green (that gates nothing and games the litmus):

1. `test\_served\_path\_runs\_on\_all\_six\_golden\_videos`: asserts `len(results) == 6`. Intent = "runs on ALL present golden videos". Change to `assert len(results) == len(_SPECS)` and keep a `>= 6` floor. Rename the test + update its docstring away from "six".
2. `test\_served\_gate\_a\_does\_not\_regress\_f1`: the ?F1 tolerance band and `mean_f1_on >= 0.5*mean_f1_off` were calibrated on the original 6 (single-court) videos; the added multicourt clips shift the aggregate. MEASURE the new mean served ?F1 on the present specs (use `backend/eval/served_gate_a.py`'s` evaluate_manifest` + `aggregate`), then set the band to bracket the NEW measured value tightly AND update the honest "measured value" comment (mirror the existing "2026-06-14: mean served ?F1 ? -0.008" style with today's date 2026-06-18 and the new number). It must stay a REAL no-regression guard ? a genuine regression must still trip it.`
3. `test\_proposal\_windowing\_reproduces\_offline\_lift`: the +0.074 / 6-of-6-wins / seg 1.68?1.01 numbers are a documented property of the ORIGINAL 6-video corpus (a root-cause control). PIN this test to the original 6 specs BY NAME ? adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles ? by filtering `\_SPECS` to those names inside the test, so it keeps reproducing the documented +0.074 finding regardless of corpus growth. Add a one-line comment explaining why it's pinned.

VERIFY (env var set so the served-gate tests RUN, not skip):

```
- `SERVED_GATE_A_MANIFEST="C:/Users/avidu/Projects/badminton-highlight-indexer/output/human_lovo_manifest.json" python -m pytest tests/test_served_gate_a_regression.py
```

-q` ? all 5 pass.
- Full gate: same env var prefix + `python run\_all\_tests.py --cov=backend --cov=training --cov-fail-under=70` (green, coverage ?70 and not regressed vs ~80.7%); `ruff check .`; `mypy backend training` ? all clean. (Other pre-existing local-only failures unrelated to this test, if any, note but don't fix.)

BRANCHING SAFETY (a sibling session shares the PARENT checkout; you're in a worktree but still branch from the remote): `git fetch origin` then `git checkout -b fix/served-gate-a-litmus-grown-corpus origin/master`. Stage EXPLICIT paths only (the one test file). Re-verify `git branch --show-current` + `git log --oneline origin/master..HEAD` lists ONLY your commit before pushing. Push and open a PR (`gh pr create --base master`) whose body states the measured new mean ?F1 and what each of the 3 fixes does. Commit message + PR body end with the repo's co-author trailer: `Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>`. Do NOT auto-merge ? it recalibrates measured litmus numbers, so leave it for owner review (same posture as PR #220).

Return: the measured new mean served ?F1, confirmation all 5 served-gate tests pass + full gate green, and the PR URL.

2. assistant (2026-06-18T08:12:21.896Z)

I'll start by reproducing the failures. Let me first pull the latest and examine the test file and the supporting eval module.

3. user (2026-06-18T08:12:26.846Z)

```
---BRANCH---  
worktree-agent-a89ebe7a124c60412  
---STATUS---
```

4. user (2026-06-18T08:12:26.910Z)

```
1      """SERVED Gate-A regression litmus ? the *production* FusionSegmenter path on real  
footage.  
2  
3      The flagged gap (#146): the served default flip (`default_segmenter=fusion_hybrid` +  
4      ``min_crossings=2``) is **mock-tested only** on real video (`test_fusion_hybrid.py`  
drives a  
5      MagicMock runner). The offline ablation litmus  
(`test_ablation.test_litmus_reproduces_gate_a`)  
6      validates the Gate-A *signal* ? but on the **permissive proposal windowing**  
7      (`distill_local.PROPOSAL`), NOT the production operating point. This module closes the  
gap: it  
8      runs the real ``FusionSegmenter`` decision seams (`_enrich_candidate_stats`` net-  
crossings +  
9      ``_select_for_handoff`` Gate A) over windows from the **production**  
10     ``trajectory_to_action_windows`` config, on the 6 cached golden WASB trajectories, and  
asserts  
11     the served path's measured behaviour.  
12  
13     GPU-free / Gemini-free (person cue OFF) ? it stops at the AI-handoff boundary. Skips  
cleanly in  
14     CI where the golden trajectories are absent; runs on the owner box where they are  
present.  
15  
16     HONEST FINDING baked into the assertions (2026-06-14): on the **production windowing**  
Gate A is  
17     NOT the +0.074 win the offline harness reports. Production windowing is already  
conservative  
18     (velocity_threshold 0.02, merge_gap 2.0, min_window 2.0) ? few long windows, over-seg  
ratio  
19     already < 1 ? there is little held-shuttle junk left for Gate A to cut. So the served  
?F1 is  
20     ~neutral (slightly negative), NOT +0.074. The +0.074 reproduces ONLY under the  
permissive
```

```

21     proposal windowing (asserted in ``test_proposal_windowing_reproduces_offline_lift``
below ? that
22     is what the offline litmus measures). These tests pin BOTH facts so a future windowing
change
23     that silently revives ? or breaks ? either one trips the litmus.
24     """
25     from __future__ import annotations
26
27     import json
28     import os
29
30     import pytest
31
32     # cv2 is imported transitively by trajectory_hybrid (the served engine). On a box
without it
33     # (CI), skip the whole module rather than erroring at import.
34     pytest.importorskip("cv2")
35
36     from backend.eval import served_gate_a as sga # noqa: E402
37     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner # noqa: E402
38     from backend.pipeline.detectors import rally_gate # noqa: E402
39     from backend.eval.calibrate_wasb import load_golden_gts # noqa: E402
40     from backend.eval.metrics import score # noqa: E402
41     from backend.eval import segmentation_metrics as _segm # noqa: E402
42
43     # The golden manifest lives in the MAIN checkout's output/ (gitignored; absent in a
fresh
44     # worktree / CI). Resolve it relative to this repo root first, with an env override for
an
45     # out-of-tree checkout.
46     _REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
47     MANIFEST = os.environ.get(
48         "SERVED_GATE_A_MANIFEST",
49         os.path.join(_REPO_ROOT, "output", "human_lovo_manifest.json"),
50     )
51
52
53     def _golden_specs():
54         """Return the manifest specs whose trajectory + label files are all present, else
[ ]."""
55         if not os.path.isfile(MANIFEST):
56             return []
57         with open(MANIFEST, encoding="utf-8") as f:
58             specs = json.load(f)
59         ok = [s for s in specs
60              if os.path.isfile(str(s.get("trajectory", "")))
61              and os.path.isfile(str(s.get("labels", "")))]
62         return ok if len(ok) >= 6 else []
63
64
65     _SPECS = _golden_specs()
66     _skip = pytest.mark.skipif(
67         not _SPECS,
68         reason=f"golden trajectories absent (manifest {MANIFEST!r}); runs on the owner box",
69     )
70
71
72     # ----- served-path
litmus
73
74     @_skip
75     def test_served_path_runs_on_all_six_golden_videos():
76         """The production FusionSegmenter decision path runs end-to-end (no GPU/Gemini) on
all 6
77         golden trajectories and yields a handoff window list for each, for both gate

```

```

settings."""
78     results = sga.evaluate_manifest(_SPECS)
79     assert len(results) == 6
80     for r in results:
81         assert r.num_gt > 0
82         assert r.n_handoff_off >= r.n_handoff_on    # Gate A only ever drops windows,
never adds
83
84
85     @_skip
86     def test_served_gate_a_does_not_regress_f1():
87         """SERVED no-regression guard: on the production windowing Gate A must not
MATERIALLY hurt
88         F1. It is ~neutral here (the held-shuttle junk the offline harness sees is already
merged
89         away by the conservative production windowing) ? so we assert the mean ?F1 sits in a
tight
90         band around 0, never a real regression. (The +0.074 lift lives under the permissive
proposal
91         windowing ? see the dedicated test below.)"""
92         results = sga.evaluate_manifest(_SPECS)
93         agg = sga.aggregate(results)
94         # Honest measured value (2026-06-14): mean served ?F1 ? -0.008. Pin a tolerance band
so a
95         # real regression (Gate A actively destroying served F1) trips, while the measured
~neutral
96         # result passes.
97         assert -0.03 <= agg["mean_delta_f1"] <= 0.03, agg
98         # And Gate A must never collapse served F1 to ~0 on the production path.
99         assert agg["mean_f1_on"] >= 0.5 * agg["mean_f1_off"] - 1e-9, agg
100
101
102     @_skip
103     def test_served_gate_a_does_not_increase_over_segmentation():
104         """Gate A only ever DROPS windows, so the served over-seg ratio must not rise when
it's on
105         (the direction-of-effect invariant ? even where the absolute lift is ~neutral)."""
106         results = sga.evaluate_manifest(_SPECS)
107         agg = sga.aggregate(results)
108         assert agg["mean_seg_ratio_on"] <= agg["mean_seg_ratio_off"] + 1e-9, agg
109         # Per-video too: the gate is monotone (kept windows ? all windows).
110         for r in results:
111             assert r.seg_ratio_on <= r.seg_ratio_off + 1e-9, r
112
113
114     @_skip
115     def test_served_uses_production_windowing_operating_point():
116         """Pin the served operating point to config.json's indexing defaults ? if someone
retunes
117         production windowing, this litmus's premise (and the divergence it documents) is
revisited
118         deliberately, not silently."""
119         assert sga.PROD_VELOCITY_THRESH == 0.02
120         assert sga.PROD_MERGE_GAP == 2.0
121         assert sga.PROD_MIN_WINDOW == 2.0
122
123
124     # ----- the +0.074 lives under PROPOSAL
windowing
125
126     # distill_local.PROPOSAL ? the permissive, recall-oriented candidate windowing the
OFFLINE
127     # ablation/fusion_compare harness builds on (low velocity thresh, short merge/min-dur ?
many
128     # small windows). The +0.074 Gate-A lift is a property of THIS windowing, not the served

```

```

one.
129     _PROPOSAL = (0.005, 1.0, 0.5)
130
131
132     @_skip
133     def test_proposal_windowing_reproduces_offline_lift():
134         """CONTROL / root-cause pin: under the OFFLINE harness's permissive proposal
windowing, the
135         SAME ``rally_gate`` net-crossing gate the served seam uses reproduces the offline
litmus
136         (?F1 ? +0.074, 6/6 wins, over-seg 1.68?1.01). This proves the served ~neutral result
is a
137         WINDOWING difference, not a broken gate ? and locks the offline number to its true
premise."""
138         d_on, d_off, seg_on, seg_off = [], [], [], []
139         for s in _SPECS:
140             fps, fw = float(s["fps"]), float(s["frame_width"])
141             pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
142             wins = TrackNetRunner.trajectory_to_action_windows(
143                 pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
144                 merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
145             iv = [(w["start"], w["end"]) for w in wins]
146             est = rally_gate.estimate_play_axis(pts)
147             gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
148             on = [(g.start, g.end) for g in gated if g.crossings >= 2]
149             gts = load_golden_gts(s["labels"])
150             d_on.append(score(on, gts).f1)
151             d_off.append(score(iv, gts).f1)
152             seg_on.append(_segm.segment_count_ratio(on, gts))
153             seg_off.append(_segm.segment_count_ratio(iv, gts))
154         n = len(d_on)
155         mean_delta = (sum(d_on) - sum(d_off)) / n
156         wins = sum(1 for a, b in zip(d_on, d_off) if a > b + 1e-9)
157         assert mean_delta == pytest.approx(0.074, abs=0.01), mean_delta
158         assert wins == 6, wins
159         assert sum(seg_off) / n == pytest.approx(1.68, abs=0.05)
160         assert sum(seg_on) / n == pytest.approx(1.01, abs=0.05)
161

```

5. assistant (2026-06-18T08:12:31.164Z)

The worktree is on its own branch with a clean status. Let me look at the eval module and verify the manifest exists before reproducing.

6. user (2026-06-18T08:12:33.092Z)

```

1     """SERVED Gate-A verification: run the *production* ``FusionSegmenter`` decision seams
2     on a cached WASB trajectory and score the resulting AI-handoff windows vs golden labels.
3
4     The offline litmus (``backend.eval.ablation`` / ``fusion_compare``) measures Gate A on
the
5     **permissive proposal windowing** (``distill_local.PROPOSAL`` = recall-oriented) re-
scored
6     through the experiment harness's ``_gate_mask``. That validates the *signal*. It does
NOT
7     exercise the **served path**: in production the windows come from
8     ``trajectory_to_action_windows`` at the production operating point
9     (``velocity_threshold=0.02, dead_time_merge_gap=2.0, min_action_window_duration=2.0``),
the
10    net-crossing count is computed by ``FusionSegmenter._enrich_candidate_stats``, and the
gate is
11    ``FusionSegmenter._select_for_handoff``. THIS module drives exactly those production
methods ?
12    no reimplementations of the windowing or the gate ? on a cached trajectory CSV, so the
13    verification reflects what ``process_video`` actually hands to Gemini.

```

```

14
15 GPU-free and Gemini-free: it stops at the handoff-selection boundary (the windows that
WOULD
16 be sent to the AI). With the person cue OFF (the default) nothing here loads torch or
touches
17 the GPU ? it just re-scores cached WASB trajectories.
18
19 Run (owner box, golden trajectories present):
20 python -m backend.eval.served_gate_a --manifest output/human_lovo_manifest.json
21 """
22 from __future__ import annotations
23
24 import argparse
25 import json
26 from dataclasses import dataclass
27 from typing import Any, Dict, List, Optional, Tuple
28
29 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
30 from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
31 from backend.eval.calibrate_wasb import load_golden_gts
32 from backend.eval.metrics import Interval, score, Score
33 from backend.eval import segmentation_metrics as _segm
34
35 # Production windowing operating point (config.json indexing defaults ? the values
36 # TrajectoryHybridSegmenter.process_video reads for the served fusion run).
37 PROD_VELOCITY_THRESH = 0.02
38 PROD_MERGE_GAP = 2.0
39 PROD_MIN_WINDOW = 2.0
40
41
42 def _served_idx_cfg(min_crossings: int) -> Dict[str, Any]:
43     """The ``indexing`` config dict the served FusionSegmenter seams read ? mirrors
44     config.json's ``indexing.fusion`` block, with Gate A at ``min_crossings`` and the
45     person cue OFF (so no GPU / torch is touched)."""
46     return {
47         "fusion": {
48             "substrate": "wasb",
49             "min_crossings": min_crossings,
50             "min_players": 0,
51             "cue_flags": {},
52         },
53         "wasb": {"velocity_threshold": PROD_VELOCITY_THRESH},
54         "dead_time_merge_gap": PROD_MERGE_GAP,
55         "min_action_window_duration": PROD_MIN_WINDOW,
56     }
57
58
59 def served_handoff_windows(trajecory_csv: str, fps: float, frame_width: float,
60                             min_crossings: int) -> List[Interval]:
61     """Run the PRODUCTION FusionSegmenter decision path on a cached trajectory and
return the
62     windows that would reach the AI handoff (i.e. the served rally candidates).
63
64     Uses the real ``FusionSegmenter`` seam methods ? ``_enrich_candidate_stats``
(computes
65     ``net_crossings``) and ``_select_for_handoff`` (Gate A) ? over windows from the
production
66     ``trajectory_to_action_windows`` operating point. No GPU, no Gemini: the person cue
is OFF.
67     """
68     points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
69     windows = TrackNetRunner.trajectory_to_action_windows(
70         points, fps=fps, frame_width=frame_width,
71         velocity_thresh=PROD_VELOCITY_THRESH, merge_gap=PROD_MERGE_GAP,
72         min_window_duration=PROD_MIN_WINDOW,

```

```

73     )
74     idx_cfg = _served_idx_cfg(min_crossings)
75     # Construct the real served segmenter (db=None, config carries the indexing block).
We only
76     # call the pure decision seems ? no process_video, no DB, no provider ? exactly as
the
77     # registry constructs it for `name`/`description`.
78     seg = FusionSegmenter(None, {"indexing": idx_cfg, "sport": "badminton"})
79     # The served per-window stats (incl. net_crossings) via the production enrichment
hook.
80     window_stats = [
81         seg._enrich_candidate_stats(cw, points, fps, frame_width, video_path="",
82                                     idx_cfg=idx_cfg)
83         for cw in windows
84     ]
85     keep = seg._select_for_handoff(windows, window_stats, idx_cfg)
86     return [(windows[i]["start"], windows[i]["end"]) for i in keep]
87
88
89     @dataclass
90     class ServedVideoResult:
91         name: str
92         num_gt: int
93         on: Score          # Gate-A ON (min_crossings=2)
94         off: Score         # Gate-A OFF (min_crossings=0, control)
95         seg_ratio_on: float # |preds| / |gts| with Gate-A ON (over-segmentation)
96         seg_ratio_off: float
97         n_handoff_on: int
98         n_handoff_off: int
99
100         @property
101         def delta_f1(self) -> float:
102             return self.on.f1 - self.off.f1
103
104
105     def evaluate_video(spec: Dict[str, Any], iou: float = 0.5,
106                       on_crossings: int = 2) -> ServedVideoResult:
107         """Score the served Gate-A ON (min_crossings=on_crossings) vs OFF (0) handoff
windows for
108         one golden video against its golden labels."""
109         fps, fw = float(spec["fps"]), float(spec["frame_width"])
110         traj, labels = str(spec["trajectory"]), str(spec["labels"])
111         gts = load_golden_gts(labels)
112         preds_on = served_handoff_windows(traj, fps, fw, min_crossings=on_crossings)
113         preds_off = served_handoff_windows(traj, fps, fw, min_crossings=0)
114         return ServedVideoResult(
115             name=str(spec["name"]),
116             num_gt=len(gts),
117             on=score(preds_on, gts, iou),
118             off=score(preds_off, gts, iou),
119             seg_ratio_on=_segm.segment_count_ratio(preds_on, gts),
120             seg_ratio_off=_segm.segment_count_ratio(preds_off, gts),
121             n_handoff_on=len(preds_on),
122             n_handoff_off=len(preds_off),
123         )
124
125
126     def evaluate_manifest(specs: List[Dict[str, Any]], iou: float = 0.5,
127                          on_crossings: int = 2) -> List[ServedVideoResult]:
128         return [evaluate_video(s, iou, on_crossings) for s in specs]
129
130
131     def aggregate(results: List[ServedVideoResult]) -> Dict[str, Any]:
132         """Mean ON/OFF F1 + over-seg ratio, the paired ?F1, and the per-video win count."""
133         n = len(results)

```

```

134
135     def _mean(getter) -> float:
136         return sum(getter(r) for r in results) / n if n else 0.0
137
138     deltas = [r.delta_f1 for r in results]
139     wins = sum(1 for d in deltas if d > 1e-9)
140     losses = sum(1 for d in deltas if d < -1e-9)
141     return {
142         "num_videos": n,
143         "mean_f1_on": _mean(lambda r: r.on.f1),
144         "mean_f1_off": _mean(lambda r: r.off.f1),
145         "mean_delta_f1": _mean(lambda r: r.delta_f1),
146         "mean_precision_on": _mean(lambda r: r.on.precision),
147         "mean_precision_off": _mean(lambda r: r.off.precision),
148         "mean_recall_on": _mean(lambda r: r.on.recall),
149         "mean_recall_off": _mean(lambda r: r.off.recall),
150         "mean_seg_ratio_on": _mean(lambda r: r.seg_ratio_on),
151         "mean_seg_ratio_off": _mean(lambda r: r.seg_ratio_off),
152         "wins": wins,
153         "losses": losses,
154     }
155
156
157     def format_table(results: List[ServedVideoResult], agg: Dict[str, Any], iou: float) ->
158     str:
159         lines = [
160             f"=== SERVED Gate-A verification (production FusionSegmenter path, IoU>={iou})
161             ===",
162             f" {'video':<24s} {'gt':>3s} {'F1_off':>7s} {'F1_on':>7s} {'?F1':>7s} "
163             f"{'segR_off':>9s} {'segR_on':>8s} {'hand_off':>9s} {'hand_on':>8s}",
164         ]
165         for r in results:
166             lines.append(
167                 f" {r.name:<24s} {r.num_gt:>3d} {r.off.f1:>7.3f} {r.on.f1:>7.3f} "
168                 f"{r.delta_f1:>7.3f} {r.seg_ratio_off:>9.2f} {r.seg_ratio_on:>8.2f} "
169                 f"{r.n_handoff_off:>9d} {r.n_handoff_on:>8d}")
170         lines += [
171             f" {'-' * 92}",
172             f" MEAN F1 {agg['mean_f1_off']:.3f} (off) -> {agg['mean_f1_on']:.3f} (on) "
173             f"?F1={agg['mean_delta_f1']:+.3f} "
174             f"P {agg['mean_precision_off']:.3f}->{agg['mean_precision_on']:.3f} "
175             f"R {agg['mean_recall_off']:.3f}->{agg['mean_recall_on']:.3f}",
176             f" over-seg ratio {agg['mean_seg_ratio_off']:.2f} (off) -> "
177             f"{agg['mean_seg_ratio_on']:.2f} (on) | wins {agg['wins']}/{agg['num_videos']}
178             "
179             f"(losses {agg['losses']})",
180         ]
181         return "\n".join(lines)
182
183
184     def run(manifest_path: str, iou: float = 0.5) -> Tuple[List[ServedVideoResult],
185     Dict[str, Any]]:
186         with open(manifest_path, encoding="utf-8") as f:
187             specs = json.load(f)
188             results = evaluate_manifest(specs, iou=iou)
189             return results, aggregate(results)
190
191
192     def main(argv: Optional[List[str]] = None) -> None:
193         ap = argparse.ArgumentParser(
194             description="Verify served Gate-A on real footage via the production
195             FusionSegmenter path.")
196         ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
197         ap.add_argument("--iou", type=float, default=0.5)
198         args = ap.parse_args(argv)

```

```

194         results, agg = run(args.manifest, iou=args.iou)
195         print(format_table(results, agg, args.iou))
196
197
198     if __name__ == "__main__":
199         main()
200

```

7. user (2026-06-18T08:12:33.791Z)

```

-rw-r--r-- 1 avidu 197609 4503 Jun 18 12:26 C:/Users/avidu/Projects/badminton-highlight-
indexer/output/human_lovo_manifest.json

```

8. assistant (2026-06-18T08:12:37.449Z)

The manifest exists. Let me inspect it to see the specs (names + which have present files), and check that cv2 is importable here.

9. user (2026-06-18T08:12:37.937Z)

```

1      [
2      {
3          "name": "adarsh_avi_singles",
4          "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv",
5          "fps": 59.94,
6          "frame_width": 1920,
7          "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh
and Avi.rallies.csv"
8      },
9      {
10         "name": "kushagra_singles",
11         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/kushagra_traj.csv",
12         "fps": 59.94,
13         "frame_width": 1920,
14         "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv"
15     },
16     {
17         "name": "largetest_doubles",
18         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/largetestvideo_traj.csv",
19         "fps": 29.97,
20         "frame_width": 1920,
21         "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv"
22     },
23     {
24         "name": "gbaaddy",
25         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/gbaaddy_traj.csv",
26         "fps": 59.94,
27         "frame_width": 1920,
28         "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/gBaaddy20May2026.rallies.csv"
29     },
30     {
31         "name": "testlarge_short",
32         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/testlarge_traj.csv",
33         "fps": 29.97,
34         "frame_width": 1920,
35         "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden

```

```

Labelled/TestLargeVideo.rallies.csv"
36     },
37     {
38         "name": "mahadevpura_singles",
39         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura_traj.csv",
40         "fps": 29.97,
41         "frame_width": 1920,
42         "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/MahadevpuraSingles.rallies.csv"
43     },
44     {
45         "name": "mahadevpura_2",
46         "fps": 59.94,
47         "frame_width": 1920.0,
48         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_2.rallies.csv",
49         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura2_june12_traj.csv"
50     },
51     {
52         "name": "GX010128",
53         "fps": 59.94,
54         "frame_width": 1920.0,
55         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010128.rallies.csv",
56         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX010128_june12_traj.csv"
57     },
58     {
59         "name": "mahadevpura_1",
60         "fps": 59.94,
61         "frame_width": 1920.0,
62         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/mahadevpura_1.rallies.csv",
63         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura1_june12_traj.csv"
64     },
65     {
66         "name": "GX030094",
67         "fps": 29.97,
68         "frame_width": 1920.0,
69         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX030094.rallies.csv",
70         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX030094_june15_traj.csv"
71     },
72     {
73         "name": "Badminton_BXH_2",
74         "fps": 59.94,
75         "frame_width": 1920.0,
76         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Badminton_BXH_2.rallies.csv",
77         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/Badminton_BXH_2_june15_traj.csv"
78     },
79     {
80         "name": "GX010141",
81         "fps": 59.94,
82         "frame_width": 1920.0,
83         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010141.rallies.csv",
84         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX010141_june17_traj.csv"
85     },

```

```

86     {
87         "name": "GX010137",
88         "fps": 59.94,
89         "frame_width": 1920.0,
90         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX010137.rallies.csv",
91         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX010137_june17_traj.csv"
92     },
93     {
94         "name": "Boxhill_Doubles",
95         "fps": 29.97,
96         "frame_width": 1920.0,
97         "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Boxhill_Doubles.rallies.csv",
98         "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/Boxhill_Doubles_june18_traj.csv"
99     },
100    {
101        "name": "GX020094",
102        "fps": 29.97,
103        "frame_width": 1920.0,
104        "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/GX020094.rallies.csv",
105        "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX020094_june18_traj.csv"
106    }
107 ]

```

10. user (2026-06-18T08:12:40.214Z)

cv2 OK 4.13.0

11. assistant (2026-06-18T08:12:42.969Z)

cv2 is importable (so the tests will run, not skip) and the manifest has 15 specs. Now let me reproduce the failures (STEP 0).

12. user (2026-06-18T08:13:09.276Z)

```

@_skip
def test_proposal_windowing_reproduces_offline_lift():
    """CONTROL / root-cause pin: under the OFFLINE harness's permissive proposal windowing,
the
    SAME ``rally_gate`` net-crossing gate the served seam uses reproduces the offline litmus
    (?F1 ? +0.074, 6/6 wins, over-seg 1.68?1.01). This proves the served ~neutral result is
a
    WINDOWING difference, not a broken gate ? and locks the offline number to its true
premise."""
    d_on, d_off, seg_on, seg_off = [], [], [], []
    for s in _SPECS:
        fps, fw = float(s["fps"]), float(s["frame_width"])
        pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
        wins = TrackNetRunner.trajectory_to_action_windows(
            pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
            merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
        iv = [(w["start"], w["end"]) for w in wins]
        est = rally_gate.estimate_play_axis(pts)
        gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
        on = [(g.start, g.end) for g in gated if g.crossings >= 2]
        gts = load_golden_gts(s["labels"])
        d_on.append(score(on, gts).f1)
        d_off.append(score(iv, gts).f1)
        seg_on.append(_segm.segment_count_ratio(on, gts))

```

```

        seg_off.append(_segm.segment_count_ratio(iv, gts))
    n = len(d_on)
    mean_delta = (sum(d_on) - sum(d_off)) / n
    wins = sum(1 for a, b in zip(d_on, d_off) if a > b + 1e-9)
> assert mean_delta == pytest.approx(0.074, abs=0.01), mean_delta
E   AssertionError: -0.00137135080069707
E   assert -0.00137135080069707 == 0.074 ± 0.01
E
E   comparison failed
E   Obtained: -0.00137135080069707
E   Expected: 0.074 ± 0.01

```

```

tests\test_served_gate_a_regression.py:157: AssertionError
===== short test summary info =====
FAILED tests/test_served_gate_a_regression.py::test_served_path_runs_on_all_six_golden_videos
FAILED tests/test_served_gate_a_regression.py::test_served_gate_a_does_not_regress_f1
FAILED tests/test_served_gate_a_regression.py::test_proposal_windowing_reproduces_offline_lift
3 failed, 2 passed in 21.18s

```

13. assistant (2026-06-18T08:13:12.878Z)

Confirmed ? exactly the 3 failures, not skips. Now let me measure the new aggregate values I need: the mean served ?F1 on all present specs (for fix #2), and the pinned-6 reproduction (for fix #3). Let me run the eval module directly.

14. user (2026-06-18T08:13:30.466Z)

```

present specs (n=9): ['mahadevpura_2', 'GX010128', 'mahadevpura_1', 'GX030094',
'Badminton_BXH_2', 'GX010141', 'GX010137', 'Boxhill_Doubles', 'GX020094']

```

```

=== SERVED Gate-A verification (production FusionSegmenter path, IoU>=0.5) ===

```

video	gt	F1_off	F1_on	?F1	segR_off	segR_on	hand_off	hand_on
mahadevpura_2	40	0.000	0.000	+0.000	0.12	0.12	5	5
GX010128	52	0.060	0.060	+0.000	0.29	0.29	15	15
mahadevpura_1	10	0.000	0.000	+0.000	0.10	0.10	1	1
GX030094	41	0.282	0.242	-0.039	0.73	0.61	30	25
Badminton_BXH_2	44	0.000	0.000	+0.000	0.09	0.09	4	4
GX010141	29	0.333	0.146	-0.187	0.66	0.41	19	12
GX010137	34	0.597	0.576	-0.021	0.97	0.74	33	25
Boxhill_Doubles	43	0.035	0.036	+0.001	0.33	0.30	14	13
GX020094	40	0.444	0.388	-0.056	1.25	0.68	50	27

```

-----
MEAN  F1 0.195 (off) -> 0.161 (on)  ?F1=-0.034  P 0.218->0.216  R 0.186->0.132
over-seg ratio 0.50 (off) -> 0.37 (on)  | wins 1/9 (losses 4)

```

```

mean_delta_f1 = -0.033639964867538945
mean_f1_on = 0.16094581911374345
mean_f1_off = 0.1945857839812824
0.5*mean_f1_off = 0.0972928919906412
mean_f1_on >= 0.5*mean_f1_off ? True
mean_seg_ratio_on = 0.3711710588246987  mean_seg_ratio_off = 0.5041577767644286

```

15. assistant (2026-06-18T08:13:35.374Z)

Important finding: the new measured mean served ?F1 on the 9 present specs is **** -0.034 ****, which falls just outside the current `[-0.03, 0.03]` band (so the test correctly trips). The added multicourt clips shift the aggregate more negative than the original 6.

Now let me measure the pinned-6 reproduction (fix #3) to confirm the +0.074 still holds when restricted to the original 6 names. Note: the original 6 are NOT currently present in this worktree's manifest resolution ? wait, they ARE in the manifest but let me check whether their trajectory/label files exist (they're the first 6 entries pointing to "Annotation Setup").

16. user (2026-06-18T08:13:40.486Z)

```
adarsh_avi_singles traj= True labels= False
kushagra_singles traj= True labels= False
largetest_doubles traj= True labels= False
gbaaddy traj= True labels= False
testlarge_short traj= True labels= False
mahadevpura_singles traj= True labels= False
```

17. assistant (2026-06-18T08:13:45.516Z)

Critical finding: the ****original 6 videos'** label files are NOT present** in the main checkout ? their `labels` paths point to `Annotation Setup/Collect/Golden Labelled/` which don't exist. That's exactly why `\_golden\_specs()` filters them out (it requires both trajectory AND labels present), leaving only the 9 newer specs whose labels live in `output/`.

This is a major problem for fix #3 as specified. The instructions say to pin `test\_proposal\_windowing\_reproduces\_offline\_lift` to the original 6 by name ? but those 6 cannot be evaluated because their golden label files are absent. Let me verify by checking what files actually exist in those directories.

18. user (2026-06-18T08:13:47.628Z)

```
ls: cannot access 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/': No such
file or directory
```

```
---TRAJ DIR---
```

```
Badminton_BXH_2_june15_traj.csv
Boxhill_Doubles_june18_traj.csv
GX010128_june12_traj.csv
GX010137_june17_traj.csv
GX010141_june17_traj.csv
GX020094_june18_traj.csv
GX030094_june15_traj.csv
adarsh_avi_full_traj.csv
adarsh_avi_traj.csv
adarsh_traj.csv
gbaaddy_traj.csv
kushagra_traj.csv
largetestvideo_traj.csv
mahadevpura1_june12_traj.csv
mahadevpura2_june12_traj.csv
mahadevpura_traj.csv
pickleball_traj.csv
roora
sample_long_traj.csv
tabletennis_traj.csv
testlarge_traj.csv
```

19. assistant (2026-06-18T08:13:50.925Z)

The "Golden Labelled" directory doesn't exist at all. The original 6 videos' trajectory CSVs are present, but their golden label files are gone. Let me search the whole machine for these label files in case they moved.

20. user (2026-06-18T08:14:44.963Z)

The user doesn't want to proceed with this tool use. The tool use was rejected (eg. if it was a file edit, the new_string was NOT written to the file). STOP what you are doing and wait for the user to tell you how to proceed.

21. user (2026-06-18T08:14:44.964Z)

[Request interrupted by user for tool use]